Statistical Learning Exercices

Exercice 1. *Assume that we have a set of data that describes the following three variables:* $Hair = \{B, D\}$, *where* $B = blond, D = dark$. $Height = \{T, S\}$, *where* $T = tall, S = short$. $Country = \{G, P\}$, *where* $G = Greenland, P = Poland$. *You are given the following workout data set (Hair, Height, Country):*

$$\begin{pmatrix} (B,T,G) & (B,T,G) & (B,T,P) \\ (D,T,G) & (D,T,G) & (B,T,P) \\ (D,T,G) & (D,T,G) & (B,T,P) \\ (D,T,G) & (D,T,G) & (D,T,P) \\ (B,T,G) & (B,T,G) & (D,T,P) \\ (B,S,G) & (B,S,G) & (D,S,P) \\ (B,S,G) & (B,S,G) & (B,S,P) \\ (D,S,G) & (D,S,G) & (D,S,P) \end{pmatrix}$$

We want to answer the following question: if you observe a new tall individual having blond hair, what is her most likely country of origin?

Solution*:*

We need to compute and compare the posterior probabilities for the two possibilities $G = Greenland, P = Poland$.

$$\mathbb{P}(G|B,T) \text{ versus } \mathbb{P}(P|B,T)$$

We have that these posterior probabilities are given by using the Bayes rule:

$$\mathbb{P}(G|B,T) = \frac{\mathbb{P}(B,T|G)\mathbb{P}(G)}{\mathbb{P}(B,T)} \tag{1}$$

Likewise

$$\mathbb{P}(P|B,T) = \frac{\mathbb{P}(B,T|P)\mathbb{P}(P)}{\mathbb{P}(B,T)} \tag{2}$$

We also know that these two probabilities to 1 *so it is enough to compute either one of them and compare it to* $1/2$.

Or alternatively we compare (1) *and* (2) *by comparing their numerators since both have the same denominator. Here we take this route (as it seems slightly shorter). We need to compute:* $\mathbb{P}(B,T|G), \mathbb{P}(G)$ *and* $\mathbb{P}(B,T|P), \mathbb{P}(P)$.

1. *Give the maximum posterior solution (MAP) to the question, using Naive Bayes (NB) assumption. Detail your work.*

   The naive Bayes assumption gives $\mathbb{P}(B,T|G) = \mathbb{P}(B|G)\mathbb{P}(T|G)$ *and* $\mathbb{P}(B,T|P) = \mathbb{P}(B|P)\mathbb{P}(T|P)$. *We plug these in the numerators of the Equations* (1) *and* (2).

   Numerator of (1) *is* $\mathbb{P}(B|G)\mathbb{P}(T|G)\mathbb{P}(G)$ *and Numerator of* (2) *is* $\mathbb{P}(B|P)\mathbb{P}(T|P)\mathbb{P}(P)$.

   Now we need to estimate $\mathbb{P}(B|G), \mathbb{P}(T|G), \mathbb{P}(G)$, *and* $\mathbb{P}(B|P), \mathbb{P}(T|P), \mathbb{P}(P)$. *We use the provided training set to estimate these probabilities.*

The estimate $\mathbb{P}(B|G)$, is the fraction of time we observe 'B' out of all those examples that belongs to the class 'G'. So we have:

$\mathbb{P}(B|G) = \frac{8}{16} = \frac{1}{2}$.

Likewise $\mathbb{P}(T|G) = \frac{10}{16} = \frac{5}{8}, \mathbb{P}(P|G) = \frac{4}{8} = \frac{1}{2}, \mathbb{P}(T|P) = \frac{5}{8}$.

The estimate of the prior probabilities $\mathbb{P}(G)$ and $\mathbb{P}(P)$ are fraction of examples that belongs to the class 'G' and 'P' respectively.

$\mathbb{P}(G) = \frac{2}{3}, \mathbb{P}(P) = \frac{1}{3}$.

Plugging these back in numerators of Equations (1) *and* (2) *we get:*

Numerator of (1) *is* $\frac{8}{16} \cdot \frac{10}{16} \cdot \frac{2}{3} = 0.2083$

Numerator of (2) *is* $\frac{4}{8} \cdot \frac{5}{8} \cdot \frac{1}{3} = 0.1042$

The Numerator of (1) *is larger than Numerator of* (2) *, then the maximum a posteriori solution (MAP) to the question, by using the NB assumption is 'Greenland'.*

2. *Give the maximum a posteriori solution (MAP) to the question, without using the naive frame from Bayes. Detail your work.*

   We need to compute the (MAP) answer without NB assumption.

   Again we compare (1) *and* (2) *by comparing their numerators. But nowe we cannot assume anything about $\mathbb{P}(B, T|G)$ and $\mathbb{P}(B, T|P)$*

   We need to estimate these joint conditional probabilities from the training set provided.

   The estimate of $\mathbb{P}(B, T|G)$ is the fraction of time we observe 'B' and 'T' out of all those examples that belongs to the class 'G'. So $\mathbb{P}(B, T|G) = \frac{2}{8}$. Likewise $\mathbb{P}(B, T|P) = \frac{3}{8}$. The prior probabilities remains unchanged so $\mathbb{P}(G) = \frac{2}{3}, \mathbb{P}(P) = \frac{1}{3}$ as before.

   Plugging these back in numerators of Equations (1) *and* (2) *we get:*

   Numerator of (1) *is* $\frac{2}{8} \cdot \frac{2}{3} = \frac{2}{24}$

   Numerator of (2) *is* $\frac{3}{8} \cdot \frac{1}{3} = \frac{3}{24}$

   The Numerator of (1) *is larger than Numerator of* (2) *, then the maximum a posteriori solution (MAP) to the question, without the NB assumption is still 'Greenland'.*

   Observe that in this question we need to estimate the joint probabilities of 2 *attributes values occurring together, I f we had not just* 2 *attributes but* 100 *attributes we would need to look for their occurrence together ans it is in general rare to find the exact configuration of such tuples occurring together. We would need a lot o training examples to have any choice to find them. This is why the NB assumption is useful approximation in such cases.*

3. *Give the maximum likelihood solution (MLE) to the question, using the NB assumption. Detail your work.*

   The MLE answer is based on comparing the likelihood terms only. So we need to compare $\mathbb{P}(B, T|G)$ versus $\mathbb{P}(B, T|P)$. In this question we use the NB assumption hence

$$\mathbb{P}(B, T|G) = \mathbb{P}(B|G)\mathbb{P}(T|G) = \frac{8}{16} \cdot \frac{10}{16} = \frac{5}{16}$$

   and

$$\mathbb{P}(B, T|P) = \mathbb{P}(B|P)\mathbb{P}(T|P) = \frac{4}{8} \cdot \frac{5}{8} = \frac{5}{16}$$

   In this case no decision can be made.

4. *Give the maximum likelihood solution (MLE) to the question, without using the NB assumption. Detail your work.*

   We need to compare the likelihoods $\mathbb{P}(B,T|G)$ versus $\mathbb{P}(B,T|P)$ without using the NB assumption.

   We have $\mathbb{P}(B,T|G) = \frac{2}{8}$ and $\mathbb{P}(B,T|P) = \frac{3}{8}$. Since $\mathbb{P}(B,T|P) > \mathbb{P}(B,T|G)$ the MLE answer without using the NB assumption is 'Poland'

Exercice 2. *We consider a tiny data set with 10 observations $(x_i, y_i)_{1\leq i\leq 10}$, with $x_i \in \mathcal{X}$ and $y_i \in \mathcal{Y} = \{-1, 1\}$. Using different machine learning algorithms, two models are built from this data set: g_1 and g_2. Recall that a model is a function from $\mathcal{X}$ to $\{-1, 1\}$. The outputs of the models on the learning data set are given by the following table:*

x_i	$g_1(x_i)$	$g_2(x_i)$	y_i
x_1	1	1	1
x_2	1	-1	1
x_3	-1	1	1
x_4	1	-1	1
x_5	1	-1	1
x_6	1	-1	-1
x_7	-1	-1	-1
x_8	-1	-1	-1
x_9	-1	1	-1
x_{10}	1	-1	-1

1. *Compute the confusion matrices of both models on the learning set.*

   Solution*:*

$g_1(p,t)$	*t=-1*	*t=1*
p=-1	*3*	*1*
p=1	*2*	*4*

$g_2(p,t)$	*t=-1*	*t=1*
p=-1	*4*	*3*
p=1	*1*	*2*

2. *We use the loss function ℓ_1 given by:*

$\ell_1(p,t)$	*t=-1*	*t=1*
p=-1	*0*	*1*
p=1	*3*	*0*

   where p is the predicted value and t the true value. Compute the empirical risk of both models on the learning set for ℓ_1.

Solution*:*

The empirical risk for g_1 *is:*

$$R_{g_1} = \frac{1}{N} \sum_{p \in \mathcal{Y}} \sum_{t \in \mathcal{Y}} g_1(p,t) \ell_1(p,t) = 0.7$$

The empirical risk for g_2 *is:*

$$R_{g_2} = \frac{1}{N} \sum_{p \in \mathcal{Y}} \sum_{t \in \mathcal{Y}} g_2(p,t) \ell_1(p,t) = 0.6$$

3. *Determine the best model based on the available information using the loss function* $L_b(p,t) = \mathbf{1}_{p \neq t}$.
Solution*:*
Recall the definition of $L_b(p,t)$*:*

$L_b(p,t)$	*t=-1*	*t=1*
p=-1	*0*	*1*
p=1	*1*	*0*

According to this loss function the empirical risk for g_1 *is:*

$$R_{g_1} = \frac{1}{N} \sum_{p \in \mathcal{Y}} \sum_{t \in \mathcal{Y}} g_1(p,t) L_b(p,t) = 0.3$$

The empirical risk for g_2 *is:*

$$R_{g_2} = \frac{1}{N} \sum_{p \in \mathcal{Y}} \sum_{t \in \mathcal{Y}} g_2(p,t) L_b(p,t) = 0.4$$